

Cambridge Crime Data Cleaning and Exploratory Data Analysis

Project Overview

This project focuses on cleaning and analyzing the Cambridge Crime Reports dataset to understand crime patterns, trends, and distributions. The goal is to transform raw crime data into meaningful insights through data cleaning and exploratory data analysis (EDA).

Dataset Description

Dataset Name: Cambridge Crime Reports Dataset

Source: Cambridge Police Department Crime Reports

Dataset Features

- File Number
- Date of Report
- Crime Date Time
- Crime
- Reporting Area
- Neighborhood
- Location

Dataset Size

- Approximately 95,923 records
- 7 columns

The dataset contains crime incidents reported in Cambridge, Massachusetts, along with information about crime type, location, neighborhood, and reporting area.

Data Cleaning Steps Performed

1. Data Loading

The dataset was loaded into a Pandas DataFrame using the `read_csv()` function.

2. Missing Value Analysis

Missing values were identified using:

```
df.isnull().sum()
```

The analysis revealed missing values in:

- Crime Date Time
- Reporting Area
- Neighborhood
- Location

3. Handling Missing Values

Rows containing missing values were removed to ensure data consistency and improve analysis quality.

```
df.dropna(inplace=True)
```

4. Duplicate Check

Duplicate records were checked using:

```
df.duplicated().sum()
```

Any duplicate records found were removed from the dataset.

5. Date-Time Processing

A major challenge was that the **Crime Date Time** column contained time ranges rather than single timestamps.

Example:

```
02/21/2009 09:20 - 09:30  
02/20/2009 22:30 - 02/21/2009 10:00
```

To perform time-based analysis, only the starting date and time were extracted.

```
df['Crime_Start'] = df['Crime Date Time'].str.split(' - ').str[0]
```

The extracted values were converted into datetime format using:

```
pd.to_datetime()
```

6. Feature Engineering

Additional columns were created for analysis:

- Year
- Month
- Day of Week

These features enabled trend analysis and seasonal pattern identification.

Exploratory Data Analysis (EDA)

Several visualizations were created to understand crime patterns.

1. Top Crime Types

A bar chart was generated to identify the most frequently occurring crimes.

Observation:

A small number of crime categories accounted for a significant proportion of reported incidents, indicating that certain crimes occur much more frequently than others.

2. Crime by Neighborhood

Neighborhood-level analysis was performed to identify areas with higher crime activity.

Observation:

Crime incidents were concentrated in specific neighborhoods, suggesting the existence of localized crime hotspots.

3. Crime Trend Over Years

A yearly trend analysis was conducted using the extracted year information.

Observation:

Reported crime incidents generally increased over time and reached higher levels around 2019–2020. This trend may reflect changes in crime patterns, reporting practices, population growth, or data collection methods.

4. Crime Distribution by Month

Monthly crime counts were analyzed to identify seasonal patterns.

Observation:

Crime activity varied across months, indicating possible seasonal influences on crime occurrence.

5. Crime Distribution by Day of Week

Crime incidents were analyzed by weekday.

Observation:

Certain days recorded higher numbers of crime reports than others, suggesting that crime activity may be influenced by weekly behavioral patterns.

Key Findings

1. Certain crime categories occur far more frequently than others.
 2. Crime incidents are concentrated in specific neighborhoods.
 3. Reported crimes generally increased over the years represented in the dataset.
 4. Crime activity varies across different months of the year.
 5. Weekly patterns exist, with some days showing higher crime frequencies.
 6. Data cleaning and preprocessing significantly improved the usability of the dataset for analysis.
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Challenges Faced

Date-Time Formatting Issues

The Crime Date Time column contained date ranges rather than single timestamps, making direct datetime conversion difficult.

Missing Values

Several records contained missing information that required cleaning before analysis.

Data Preparation

Additional preprocessing and feature engineering were required to create meaningful time-based variables such as Year and Month.

Conclusion

The Cambridge Crime Reports dataset was successfully cleaned and analyzed using Python, Pandas, NumPy, Matplotlib, and Seaborn. Missing values and formatting issues were addressed, and exploratory data analysis revealed meaningful patterns related to crime type, location, and time. The findings demonstrate how data cleaning and EDA can transform raw crime data into valuable insights that may support decision-making and public safety initiatives.

Tools and Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook